

The most expensive fifteen minutes in a factory

Aditya Swaroop, for Powertown's Machine Learning Engineer, Intern (Term-Time) application. October 2026.

A Powerblock earns its keep in one 15-minute interval a month, plus a few dozen summer afternoons when the grid calls. I built an open instrument around that fact. It finds the Massachusetts buildings where a 250 kW / 522 kWh battery pays, prices each one under the real 2026 National Grid and Eversource tariffs, and runs the battery without seeing the future. Every result is reproducible and audited.

<https://as567-code.github.io/fifteen/>

Interactive page. Value your own Green Button data in the browser, replay any site with any controller, and read the audit trail.

Four findings

78% of the ceiling

A forecast-driven controller kept this share of the perfect-foresight value on 154 real metered customers in a year no model had seen. A tuned rule-of-thumb kept 58%; a fixed daily window kept 8%.

\$44,899 vs \$14,133

Average National Grid building with ConnectedSolutions Daily Dispatch stacked on demand savings, vs demand savings alone. Co-optimizing the two gives up only 13% of demand savings (5% under Eversource).

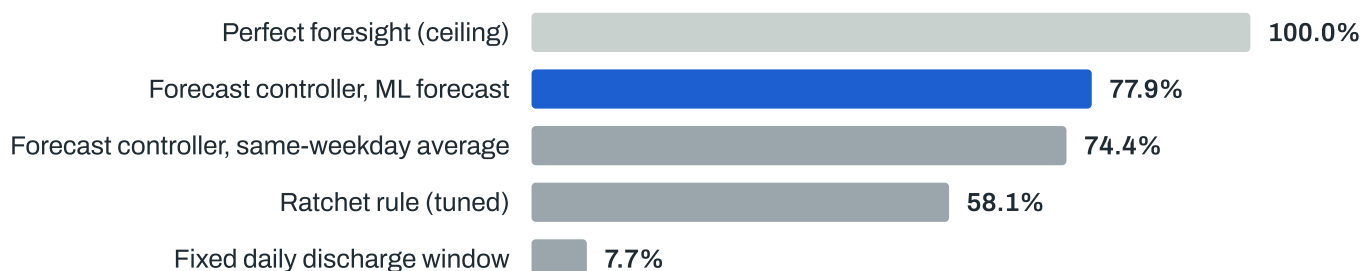
1.2 vs 0.4 months off the bill

Median library building under Eversource G-2 Boston vs National Grid G-3 (\$41,587 vs \$12,217 a year). National Grid bills transmission per kWh, which shaving cannot touch, so territory matters more than building type.

Load factor predicts almost nothing

Across 958 sites, load factor barely ranks a Powerblock's value. The energy in the top 250 kW of the hardest day's load does (rank correlation -0.84). A bill cannot show it; 15-minute data can. A twelve-bill screener still triages prospects with 14% median error on metered sites.

Controller performance, 2014 test year (Eversource G-2 billing rule)



Share of the oracle's billed-kW reduction kept, summed over 1,848 site-months. Model trained on 2012, every controller tuned on 2013 (baselines too), 2014 run once. The forecast controller never billed a month above the no-battery bill.

How it works

- **Exact perfect-foresight dispatch.** A greedy dispatch rule plus bisection finds the lowest achievable billed demand each month. It handles National Grid's peak window and Eversource's 55% off-peak discount, and it is fast enough to run in a browser. An independent HiGHS linear program confirms it to within 0.05 kW.
- **Probabilistic forecasting.** One gradient-boosted quantile model is trained across sites on scale-free load, so a new site with four weeks of data has a forecast on day one. Its 95% band covered 94.5% of actual 2014 intervals.
- **Sunk-peak control.** The controller re-plans every 15 minutes against today's forecast, corrected by the meter. It never defends below the month's already-set peak, since shaving below it is worth nothing and drains energy the hardest day will need, and it never recharges above it, so it cannot create a peak of its own.
- **Value stacking.** A month-by-month LP co-optimizes demand charges with \$200/kW ConnectedSolutions Daily Dispatch. Events go on ISO-NE's 42 highest-load summer days of 2018.

Audit trail (use AI aggressively, then audit it ruthlessly)

Check	Result
Fast optimizer vs independent LP; battery physics on every run	Agree; zero violations across 1,990 annual runs
Browser engine vs Python engine	Identical traces (1e-6 kW) and bills (to the cent)
Forecaster look-ahead; train/tune/test separation	Perturbation test passes; 2012 / 2013 / 2014
AI-gathered tariffs vs the official PDFs (hash-pinned)	Every modeled number appears verbatim in its source
Data defects found	Daylight-saving spikes (UCI), standard-time clock (ComStock), a duplicated meter (KIT): fixed and documented

What I would build first at Powertown

- Instant, audited value on every prospect, from Green Button data or twelve bills, inside the site-prospecting flow.
- A fleet controller that decides ConnectedSolutions participation per site, per day, from ISO-NE forecasts and each member's peak risk.
- Pricing National Grid's transmission coincident-peak option (\$21.91/kW at ISO-NE's monthly peak hour), a rate switch that only makes sense with a battery and a forecast.
- A tariff-watching agent whose output only merges when a check like the one above finds every number in the source PDF.

About me

M.S. Computer Science, UT Dallas (Dec 2026). Previously: GPU and distributed systems at Princeton Research Computing, fault-tolerant production services, and evaluation harnesses that grade hundreds of submissions. I built this with AI agents in the loop and my own judgment on what to trust; code, data sources and tests are in the repository linked from the page.

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